

## Change of Coordinate Matrix in $\mathbb{R}^n$

We call the “standard basis”  $\mathcal{S} = \{\mathbf{e}_1, \dots, \mathbf{e}_n\}$  and assume we have  $\mathcal{B} = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$  as a basis of our vector space  $V$ , then we defined

$$P(\mathcal{B})_{\mathcal{S}} = \begin{bmatrix} [\mathbf{b}_1]_{\mathcal{S}} & \dots & [\mathbf{b}_n]_{\mathcal{S}} \end{bmatrix}.$$

The above represents our basis vectors as an array of real numbers.

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This means if  $\mathbf{x}$  is an array of coordinates for the  $\mathcal{B}$  basis, then the output is a vector written in the standard basis!

## Change of Coordinate Matrix in $\mathbb{R}^n$ Example

Consider the basis

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} \right\}$$

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# Change of Coordinate Matrix in $\mathbb{R}^n$ is a Linear Transformation

## Theorem

*If  $V = \mathbb{R}^n$ , then the Coordinate Matrix  $P(\mathcal{B})_S$  is a Linear Transformation given by*

$$P(\mathcal{B})_S : [\mathbf{x}]_{\mathcal{B}} \mapsto \mathbf{x},$$

*which changes a vector from  $\mathcal{B}$  coordinates to being written in the standard basis*

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*which changes a vector from  $\mathcal{B}$  coordinates to being written in the standard basis and*

$$P(\mathcal{B})_{\mathcal{S}}^{-1} : \mathbf{x} \mapsto [\mathbf{x}]_{\mathcal{B}}$$

*changes a vector from the standard basis back to  $\mathcal{B}$  coordinates.*

# Changing Coordinate From One Basis to Another

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*Let  $\mathcal{B} = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$  and  $\mathcal{C} = \{\mathbf{c}_1, \dots, \mathbf{c}_n\}$  be bases for a vector space  $V$ .*

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Is there a potentially better way?

# A Better Way Without Explicitly Computing an Inverse

Yes!

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We also have that

$$P_{\mathcal{C} \leftarrow \mathcal{B}} = P(\mathcal{B})_{\mathcal{C}} = \begin{bmatrix} [\mathbf{b}_1]_{\mathcal{C}} & \cdots & [\mathbf{b}_n]_{\mathcal{C}} \end{bmatrix}$$

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Which we can solve in one of two ways

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2) Solve

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Which is solving the equation

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then plugging in  $\mathbf{y} = \mathbf{b}_1, \dots, \mathbf{b}_n$ .

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Which is solving the equation

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then plugging in  $\mathbf{y} = \mathbf{b}_1, \dots, \mathbf{b}_n$ .

Either will give the right answer, it's just a matter of preference!

## Change of Coordinate Example

Consider the following bases of  $\mathbb{R}^3$ .

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} \right\}, \mathcal{C} = \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

Compute  $P_{\mathcal{C} \leftarrow \mathcal{B}}$ .

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Compute  $P_{\mathcal{C} \leftarrow \mathcal{B}}$ . First, we try method 1.

$$\left[ P(\mathcal{C})_{\mathcal{S}} \mid P(\mathcal{B})_{\mathcal{S}} \right] = \left[ \begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 2 \\ 1 & 1 & 0 & 2 & 1 & 1 \\ 1 & 1 & 1 & 1 & 2 & 0 \end{array} \right]$$

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$$\left[ P(\mathcal{C})_{\mathcal{S}} \mid P(\mathcal{B})_{\mathcal{S}} \right] = \left[ \begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 2 \\ 1 & 1 & 0 & 2 & 1 & 1 \\ 1 & 1 & 1 & 1 & 2 & 0 \end{array} \right] \rightarrow \left[ \begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 2 \\ 0 & 1 & 0 & 1 & 1 & -1 \\ 0 & 1 & 1 & 0 & 2 & -2 \end{array} \right]$$

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So

$$P_{\mathcal{C} \leftarrow \mathcal{B}} = \begin{bmatrix} 1 & 0 & 2 \\ 1 & 1 & -1 \\ -1 & 1 & -1 \end{bmatrix}$$

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Compute  $P_{\mathcal{C} \leftarrow \mathcal{B}}$ . Next, we try method 2.

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So the columns of  $P_{\mathcal{C} \leftarrow \mathcal{B}}$  are given by

$$[\mathbf{b}_1]_{\mathcal{C}} = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}, [\mathbf{b}_2]_{\mathcal{C}} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, [\mathbf{b}_3]_{\mathcal{C}} = \begin{bmatrix} 2 \\ -1 \\ -1 \end{bmatrix}$$

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## Change of Coordinate Practice

Consider the following bases of  $\mathbb{R}^3$ .

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}, \mathcal{C} = \left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} \right\}$$

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$$P_{\mathcal{C} \leftarrow \mathcal{B}} = \begin{bmatrix} 0 & \frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{1}{4} & \frac{3}{4} \\ \frac{1}{2} & -\frac{1}{4} & \frac{1}{4} \end{bmatrix}$$

# Matrix Vector Space Practice

Let  $V$  be the vector space of **symmetric  $2 \times 2$**  real matrices. We have the basis

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}.$$

Consider the set  $\mathcal{C} = \left\{ \begin{bmatrix} 1 & -2 \\ -2 & 2 \end{bmatrix}, \begin{bmatrix} 0 & 2 \\ 2 & 4 \end{bmatrix}, \begin{bmatrix} 3 & -1 \\ -1 & 1 \end{bmatrix} \right\}$ . Answer the following questions

- 1) Show that  $\mathcal{C}$  is a basis for  $V$ .
- 2) Write the matrix corresponding to  $\begin{bmatrix} \begin{bmatrix} 2 \\ 3 \\ -2 \end{bmatrix} \end{bmatrix}_{\mathcal{C}}$  in  $\mathcal{B}$  coordinates

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Set up and show

$$P_{\mathcal{B} \leftarrow \mathcal{C}} = \begin{bmatrix} 1 & 0 & 3 \\ -2 & 2 & -1 \\ 2 & 4 & 1 \end{bmatrix}$$

2) Write the matrix corresponding to

$$\left[ \begin{bmatrix} 2 \\ 3 \\ -2 \end{bmatrix} \right]_{\mathcal{C}} \text{ in } \mathcal{B} \text{ coordinates}$$

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$$\begin{bmatrix} -4 & 4 \\ 4 & 14 \end{bmatrix}$$